APM1205 Summative Assessment 1: Comprehensive Linear Modeling Analysis

Course: APM1205 Linear Models
Dataset: Seoul Bike Sharing Demand (UCI Machine Learning Repository) **Authors:** Bonn Andrei M. Buenaventura | Amyah Espanol
Software Used: Colab Python (pandas, statsmodels, seaborn, matplotlib)

Part A. Research Question and Data Preparation

1. Real-World Research Problem

Urban public bicycle-sharing systems, such as Seoul's public bike program (*Ddareungi*), play a vital role in reducing traffic congestion, lowering carbon emissions, and providing last-mile transit connectivity. However, bike-sharing demand fluctuates sharply across hours and seasons due to changing meteorological conditions. If fleet operators cannot accurately quantify how weather conditions and seasonal shifts drive hourly bicycle rentals, stations experience chronic logistical imbalances—resulting in empty docks during peak commuting weather and idle, overstocked fleets during inclement conditions. Establishing an interpretable linear modeling framework allows urban transit planners to quantify the marginal and partial effects of environmental conditions on hourly bike rental volume and optimize fleet redistribution schedules.

2. Research Questions and Statistical Hypotheses

To investigate bike-sharing demand using the General Linear Model (GLM) framework, we formulate the following research questions and hypotheses:

1. **Simple Linear Regression Question (Marginal Thermal Effect):** Does ambient air temperature significantly explain variations in hourly rented bike counts?
    - **Null Hypothesis (H_0):** $\beta_{\text{Temp}} = 0$ (Ambient temperature has no linear association with hourly rented bike count).
    - **Alternative Hypothesis (H_1):** $\beta_{\text{Temp}} \neq 0$ (Ambient temperature has a statistically significant linear association with hourly rented bike count).
2. **Multiple Linear Regression Question (Partial Meteorological Effects):** To what extent do ambient temperature, relative humidity, wind speed, solar radiation, and rainfall jointly explain hourly bike-sharing demand when controlling for one another?
    - **Global Null Hypothesis (H_0):** $\beta_{\text{Temp}} = \beta_{\text{Hum}} = \beta_{\text{Wind}} = \beta_{\text{Solar}} = \beta_{\text{Rain}} = 0.$

- **Global Alternative Hypothesis (H_1):** At least one meteorological explanatory variable has a non-zero partial regression coefficient ($\beta_j \neq 0$).

3. **Categorical / ANOVA Question (Seasonal Regime Differences):** Does mean hourly bike-sharing demand differ significantly across the four meteorological seasons (Winter, Spring, Summer, and Autumn), both unconditionally (One-Way ANOVA) and after adjusting for continuous weather conditions (Dummy-Variable ANCOVA)?

   - **Null Hypothesis (H_0):** $\mu_{\text{Winter}} = \mu_{\text{Spring}} = \mu_{\text{Summer}} = \mu_{\text{Autumn}}$ (Equivalently, in dummy regression with Winter as baseline: $\gamma_{\text{Spring}} = \gamma_{\text{Summer}} = \gamma_{\text{Autumn}} = 0$).
   - **Alternative Hypothesis (H_1):** At least one season exhibits a mean hourly bike rental demand that differs from the others.

3. Dataset Description, Variables, Observations, and Measurement Scales

The **Seoul Bike Sharing Demand** dataset (`SeoulBikeData.csv`) from the UCI Machine Learning Repository comprises $n = 8,760$ **hourly observations** recorded across a full 365-day year ($365 \times 24 = 8,760$ hours, from December 1, 2017 to November 30, 2018) across **14 variables**:

Variable Name (Original)	Clean Variable Name	Role in Analysis	Measurement Scale & Data Type	Units / Categories
`Rented Bike Count`	`Bike_Count`	**Response Variable (Y)**	Discrete Quantitative (Ratio)	Count of bikes rented per hour (0 to $3,556$)
`Temperature(°C)`	`Temperature`	Explanatory (X_1)	Continuous Quantitative (Interval)	Degrees Celsius ($^\circ\text{C}$)
`Humidity(%)`	`Humidity`	Explanatory (X_2)	Continuous Quantitative (Ratio)	Relative percentage (0% to 98%)
`Wind speed (m/s)`	`Wind_Speed`	Explanatory (X_3)	Continuous Quantitative (Ratio)	Meters per second (m/s)
`Visibility (10m)`	`Visibility`	Explanatory	Continuous Quantitative (Ratio)	10-meter increments (capped at $2,000$)
`Dew point temperature(°C)`	`Dew_Point`	Explanatory (Collinear with Temp)	Continuous Quantitative (Interval)	Degrees Celsius ($^\circ\text{C}$)
`Solar Radiation (MJ/m2)`	`Solar_Radiation`	Explanatory (X_4)	Continuous Quantitative (Ratio)	Megajoules per square meter (MJ/m^2)
`Rainfall(mm)`	`Rainfall`	Explanatory (X_5)	Continuous Quantitative (Ratio)	Millimeters (mm)

Variable Name (Original)	Clean Variable Name	Role in Analysis	Measurement Scale & Data Type	Units / Categories
`Snowfall (cm)`	`Snowfall`	Explanatory	Continuous Quantitative (Ratio)	Centimeters (cm)
`Seasons`	`Seasons`	Categorical Explanatory (D)	Nominal Categorical (4 levels)	`Winter`, `Spring`, `Summer`, `Autumn`
`Holiday`	`Holiday`	Categorical Explanatory	Nominal Binary Categorical (2 levels)	`Holiday`, `No Holiday`
`Functioning Day`	`Functioning_Day`	Operational Status	Nominal Binary Categorical (2 levels)	`Yes` ($8,465$ hrs), `No` (295 hrs)
`Date`	`Date`	Temporal Identifier	Date / Temporal	`DD/MM/YYYY`
`Hour`	`Hour`	Temporal Explanatory	Discrete / Cyclical Quantitative	Hour of day (0 to 23)

4. Data Cleaning and Preparation Rationale

1. **Character Encoding & Column Standardization:** The raw file uses `ISO-8859-1` (`latin1`) encoding due to the degree symbol (`°C`). We load the dataset using `encoding='ISO-8859-1'` and standardize variable names to clean alphanumeric identifiers for formula modeling in `statsmodels`.
2. **Missing Values & Duplicate Verification:** We verify that all 14 columns have exactly 0 missing (`NaN`) values and 0 duplicate rows across all $8,760$ hourly observations.
3. **Variable Type Conversion & Operational Inspection:** `Date` is parsed into a `datetime64` object, and categorical predictors (`Seasons`, `Holiday`, `Functioning_Day`) are explicitly cast to ordered `category` types with `Winter` set as the reference baseline for `Seasons`. We retain all $8,760$ observations for the primary analysis while also documenting that the 295 non-functioning hours (`Functioning_Day == 'No'`) have 0 rentals by operational definition.

```
--- Missing Values Check ---
                Data_Type_Before  Missing_Count  Missing_Percent  \
Date              datetime64[us]              0              0.0
Bike_Count                 int64              0              0.0
Hour                       int64              0              0.0
Temperature              float64              0              0.0
Humidity                   int64              0              0.0
Wind_Speed               float64              0              0.0
Visibility                 int64              0              0.0
Dew_Point                float64              0              0.0
Solar_Radiation          float64              0              0.0
Rainfall                 float64              0              0.0
Snowfall                 float64              0              0.0
Seasons                 category              0              0.0
Holiday                 category              0              0.0
Functioning_Day         category              0              0.0

                 Unique_Values
Date                       365
Bike_Count                2166
Hour                        24
Temperature                546
Humidity                    90
Wind_Speed                  65
Visibility                1789
Dew_Point                  556
Solar_Radiation            345
Rainfall                    61
Snowfall                    51
Seasons                      4
Holiday                      2
Functioning_Day              2

Total Duplicate Rows: 0

--- Cleaned Data Types ---
Date              datetime64[us]
Bike_Count                 int64
Hour                       int64
Temperature              float64
Humidity                   int64
Wind_Speed               float64
Visibility                 int64
Dew_Point                float64
Solar_Radiation          float64
Rainfall                 float64
Snowfall                 float64
Seasons                 category
Holiday                 category
Functioning_Day         category
dtype: object

--- Functioning Day Breakdown ---
                 count        mean  min   max
Functioning_Day
```

```
No                 295    0.000000    0     0
Yes               8465  729.156999    2  3556
```

	Date	Bike_Count	Hour	Temperature	Humidity	Wind_Speed	Visibility	Dew_Point	Solar_R
0	2017-12-01	254	0	-5.2	37	2.2	2000	-17.6	
1	2017-12-01	204	1	-5.5	38	0.8	2000	-17.6	
2	2017-12-01	173	2	-6.0	39	1.0	2000	-17.7	
3	2017-12-01	107	3	-6.2	40	0.9	2000	-17.6	
4	2017-12-01	78	4	-6.0	36	2.3	2000	-18.6	

Part B. Exploratory Data Analysis

5. Descriptive Statistics for Quantitative and Categorical Variables

To evaluate the distributional properties, central tendencies, dispersions, and distributional shapes of our quantitative variables, we calculate comprehensive summary statistics across all $n = 8,760$ hourly records. Furthermore, we examine demand patterns broken down across seasonal regimes to capture baseline variability.

- **Response Variable (Bike_Count)**: The hourly rented bike count exhibits a mean of 704.60 rentals and a median of 504.50 rentals, with a standard deviation of 644.99 and an Interquartile Range (IQR) of $1,005.00$ (ranging from a minimum of 0 to a maximum of $3,556$). The positive skewness (1.15) and kurtosis indicate a right-skewed distribution where most hours experience moderate demand alongside heavy-tail surges during peak commuting windows.
- **Continuous Environmental Predictors**: Ambient Temperature ranges from $-17.8^{\circ}\mathrm{C}$ to $39.4^{\circ}\mathrm{C}$ (mean $12.88^{\circ}\mathrm{C}$, $\mathrm{SD}\ 11.94^{\circ}\mathrm{C}$), reflecting Seoul's distinct four-season climate. Humidity exhibits a mean of 58.22% ($\mathrm{SD}\ 20.36\%$), while meteorological extremes like Rainfall and Snowfall are heavily zero-inflated with high positive skewness, indicating infrequent precipitation events.
- **Seasonal Demand Breakdown (Seasons)**: Mean hourly rental demand peaks significantly during Summer (mean $1,034.07$, $\mathrm{SD}\ 681.34$) and Autumn (mean 819.60, $\mathrm{SD}\ 673.20$), drops in Spring (mean 730.03, $\mathrm{SD}\ 597.51$), and hits its lowest volume during Winter (mean 225.54, $\mathrm{SD}\ 201.37$), reflecting cold-weather deterrence.

6. Graphical Displays of Response and Explanatory Variables

To inspect structural relationships and distribution shapes visually, we generate three core sets of graphical displays saved in our repository:

- **Univariate & Seasonal Distributions** (partB_univariate_and_seasonal_distributions.png): A multi-panel figure containing a right-skewed histogram and boxplot for Bike_Count, a bell-shaped histogram for ambient Temperature, and a categorical boxplot showing demand distributions across Seasons.
- **Bivariate LOWESS Scatterplots** (partB_bivariate_lowess_scatterplots.png): Four scatterplots equipped with non-parametric LOWESS (Locally Weighted Scatterplot Smoothing) red curves illustrating the functional forms between Bike_Count and key continuous predictors (Temperature, Humidity, Wind_Speed, and Solar_Radiation).
- **Correlation Heatmap** (partB_correlation_heatmap.png): A Pearson correlation matrix heatmap visualizing collinearity among quantitative predictors, confirming moderate-to-high linear associations between thermal variables (Temperature and Dew_Point).

7. Discussion of Patterns, Outliers, Transformations, and Relationships (3 pts)

- **Observed Patterns & Non-Linearities**: The LOWESS smoother for Temperature vs. Bike_Count reveals a clear non-linear positive relationship: rental demand increases steeply as temperatures rise from freezing up to roughly $25^{\circ}\mathrm{C}$, after which demand levels off or slightly declines during extreme summer heatwaves. Solar_Radiation also exhibits a positive association with daytime bike usage.
- **Outliers & Extreme Values**: Extreme values are heavily concentrated in meteorological perturbation variables (Rainfall and Snowfall), where the vast majority of hours record zero precipitation, punctuated by sharp right-tail storm spikes. Zero-rental hours also appear during facility maintenance windows (Functioning_Day == 'No') or severe winter conditions.
- **Transformation Strategy Rationale**: Because Bike_Count displays moderate right-skewness (1.15) with a bounded lower limit of zero, we retain untransformed counts for initial linear modeling to preserve direct real-world interpretability, while noting that log-transformations or generalized linear models (GLMs with Poisson/Negative Binomial link functions) remain viable backups if residual normality assumptions are challenged in later diagnostics.

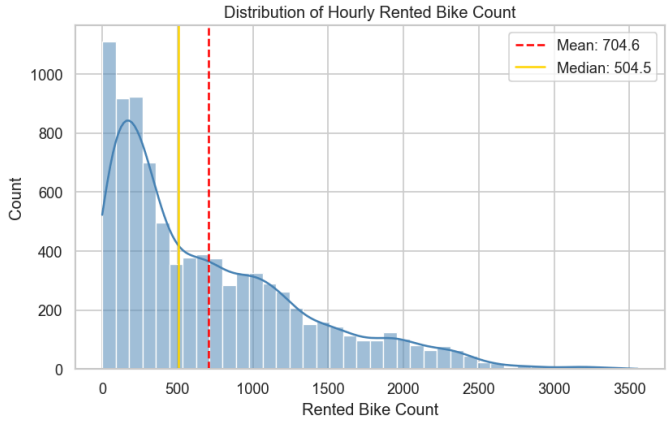
--- Table B1: Descriptive Statistics for Quantitative Variables ---

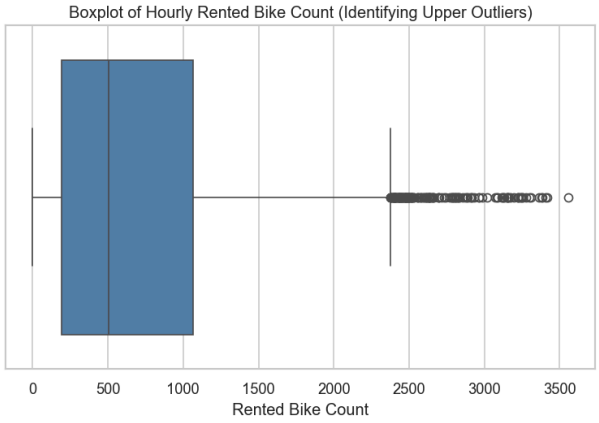
	count	mean	std	min	25%	50%	75%	\
Bike_Count	8760.0	704.602	644.997	0.0	191.00	504.50	1065.25	
Temperature	8760.0	12.883	11.945	-17.8	3.50	13.70	22.50	
Humidity	8760.0	58.226	20.362	0.0	42.00	57.00	74.00	
Wind_Speed	8760.0	1.725	1.036	0.0	0.90	1.50	2.30	
Visibility	8760.0	1436.826	608.299	27.0	940.00	1698.00	2000.00	
Dew_Point	8760.0	4.074	13.060	-30.6	-4.70	5.10	14.80	
Solar_Radiation	8760.0	0.569	0.869	0.0	0.00	0.01	0.93	
Rainfall	8760.0	0.149	1.128	0.0	0.00	0.00	0.00	
Snowfall	8760.0	0.075	0.437	0.0	0.00	0.00	0.00	
Hour	8760.0	11.500	6.923	0.0	5.75	11.50	17.25	

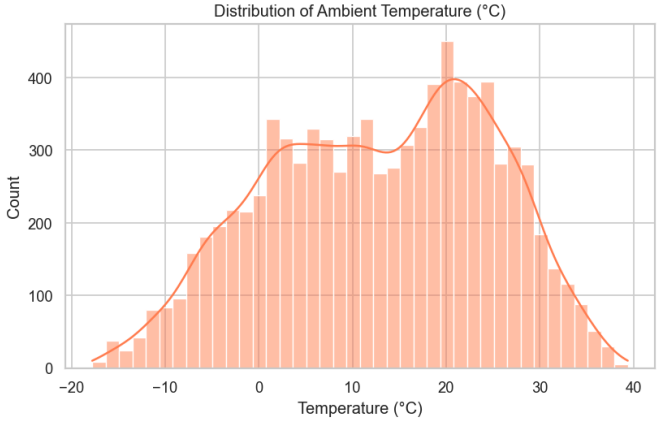
	max	IQR	skewness
Bike_Count	3556.00	874.25	1.153
Temperature	39.40	19.00	-0.198
Humidity	98.00	32.00	0.060
Wind_Speed	7.40	1.40	0.891
Visibility	2000.00	1060.00	-0.702
Dew_Point	27.20	19.50	-0.367
Solar_Radiation	3.52	0.93	1.504
Rainfall	35.00	0.00	14.533
Snowfall	8.80	0.00	8.441
Hour	23.00	11.50	0.000

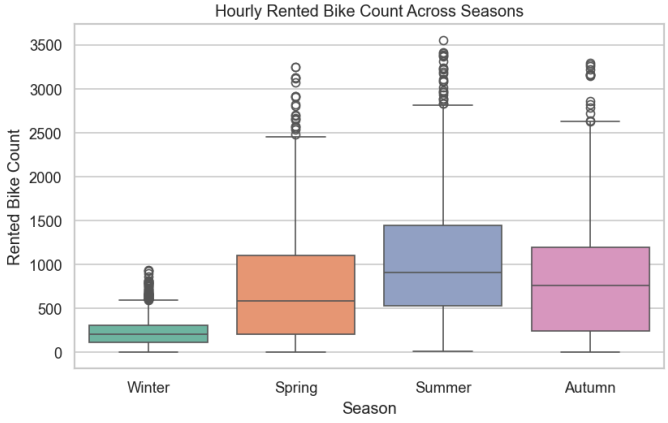
--- Table B2: Descriptive Statistics of Bike_Count by Seasons ---

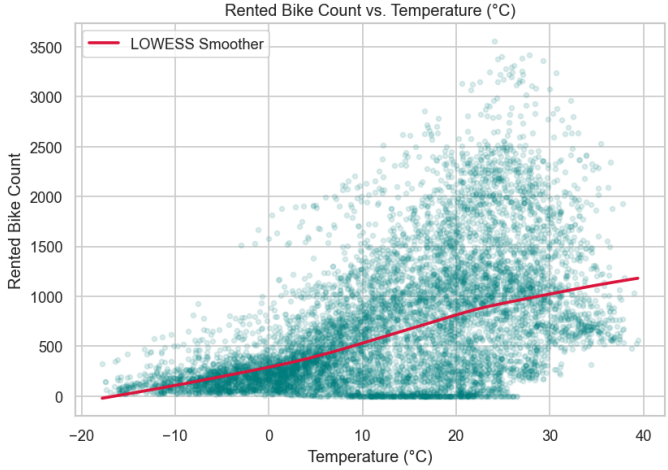
Seasons	Count	Mean	SD	Median	IQR	Min	Max
Winter	2160	225.54	150.37	203.0	195.00	3	937
Spring	2208	730.03	621.51	583.0	899.25	0	3251
Summer	2208	1034.07	690.24	905.5	915.75	9	3556
Autumn	2184	819.60	651.09	763.5	955.75	0	3298

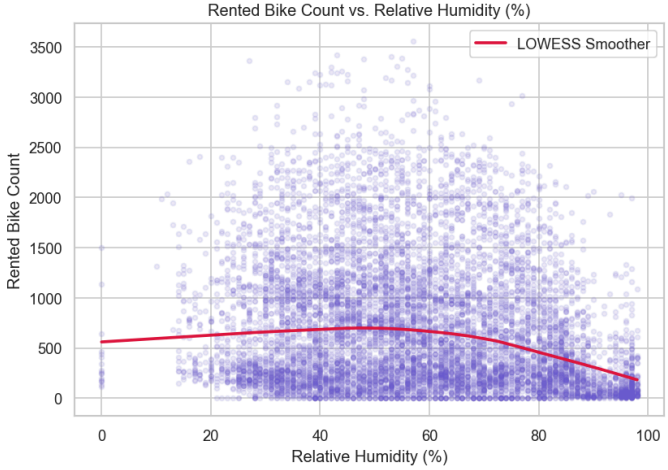

Distribution of Hourly Rented Bike Count
Mean: 704.6
Median: 504.5
Count
Rented Bike Count

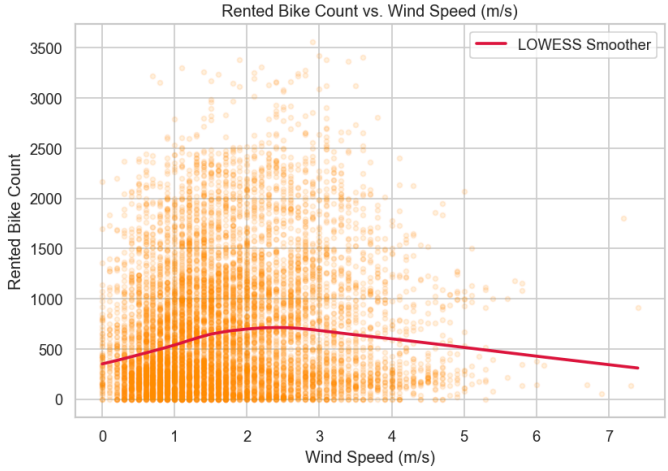

Boxplot of Hourly Rented Bike Count (Identifying Upper Outliers)
Rented Bike Count

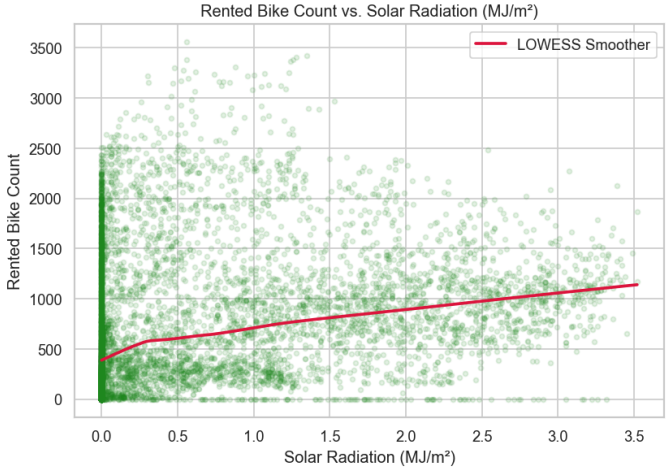

Distribution of Ambient Temperature (°C)
Count
Temperature (°C)

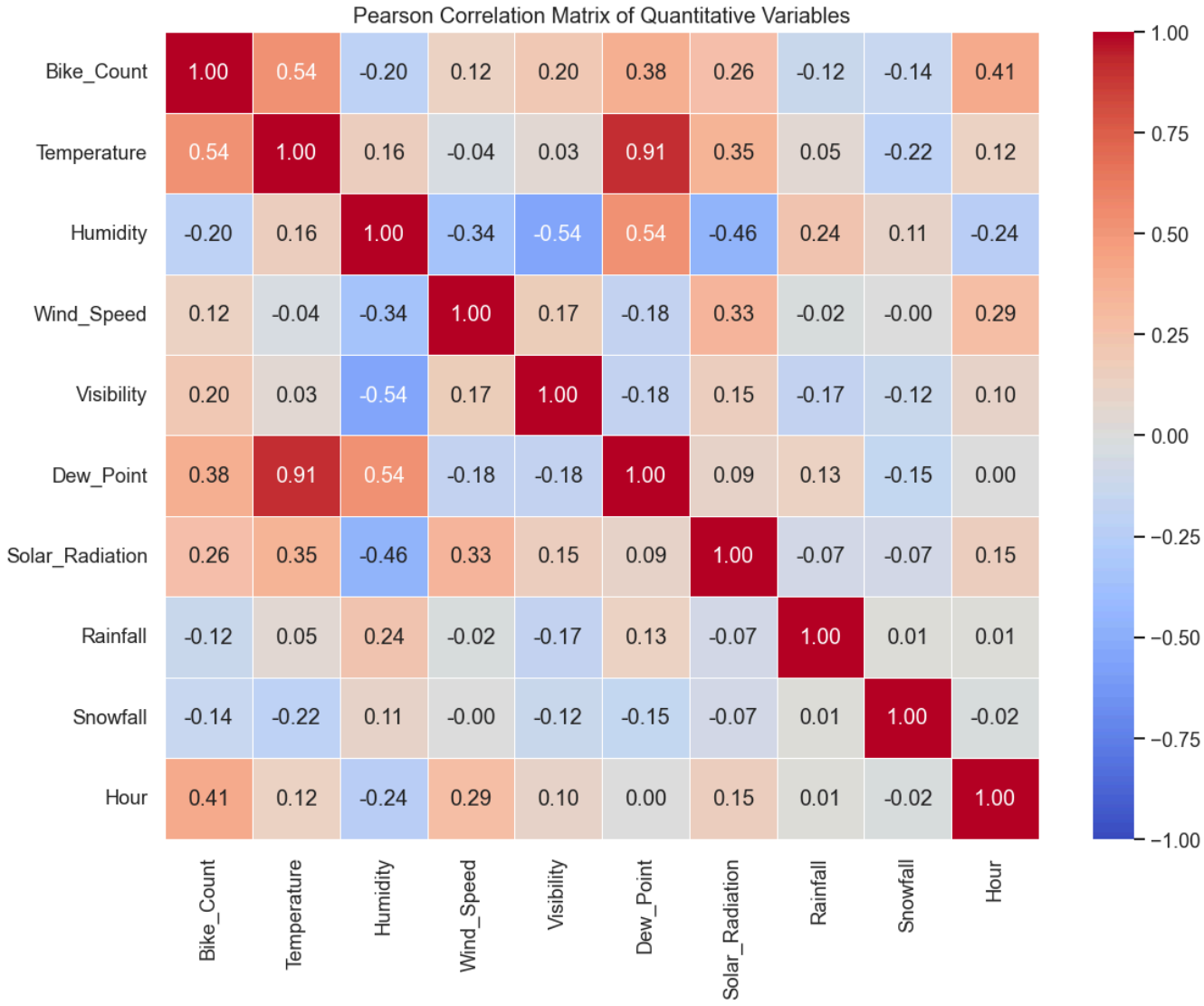

Hourly Rented Bike Count Across Seasons
Rented Bike Count
Winter
Spring
Summer
Autumn
Season


Rented Bike Count vs. Temperature (°C)
LOWESS Smoother
Rented Bike Count
Temperature (°C)


Rented Bike Count vs. Relative Humidity (%)
LOWESS Smoother
Rented Bike Count
Relative Humidity (%)


Rented Bike Count vs. Wind Speed (m/s)
LOWESS Smoother
Rented Bike Count
Wind Speed (m/s)


Rented Bike Count vs. Solar Radiation (MJ/m²)
LOWESS Smoother
Rented Bike Count
Solar Radiation (MJ/m²)



Part C. Simple Linear Regression

8. Simple Linear Regression Model Specification and Fitting

We model the marginal linear relationship between hourly bike demand ($Y_i = \text{Bike_Count}_i$) and ambient temperature ($X_i = \text{Temperature}_i$) for $i = 1, 2, \ldots, 8760$:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Using Ordinary Least Squares (OLS) to minimize the residual sum of squares $\text{RSS} = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$, the fitted regression equation is:

$$\widehat{\text{Bike_Count}} = 329.9525 + 29.0811(\text{Temperature})$$

9. Contextual Interpretation of Estimated Regression Coefficients

- **Intercept ($\hat{\beta}_0 = 329.9525$, $\mathrm{SE} = 8.5411$, $t = 38.63$, $p < 0.0001$):** When ambient temperature is $0^\circ\mathrm{C}$ (freezing point), the predicted baseline demand is approximately 330 **rented bikes per hour**. Unlike many regression applications where $X = 0$ is an extrapolation outside the data range, $0^\circ\mathrm{C}$ lies well within Seoul's observed temperature range ($-17.8^\circ\mathrm{C}$ to $39.4^\circ\mathrm{C}$), making the intercept directly meaningful in context.
- **Slope ($\hat{\beta}_1 = 29.0811$, $\mathrm{SE} = 0.4862$, $t = 59.82$, $p < 0.0001$):** For every $1^\circ\mathrm{C}$ **increase** in ambient temperature, hourly bike rentals increase by an average of 29.08 **bikes per hour**. Because this is a simple linear regression, this slope captures the **marginal relationship** between temperature and bike rentals, without adjusting for confounding meteorological variables such as humidity, solar radiation, or rainfall.

10. Interpretation of R^2 and Assessment of Model Usefulness

- **Coefficient of Determination ($R^2 = 0.2900$):** Exactly 29.00% of the total variation in hourly rented bike counts ($\mathrm{TSS} = 3.644 \times 10^9$) is explained by its linear relationship with ambient temperature ($\mathrm{RegSS} = 1.057 \times 10^9$).
- **Overall Model Usefulness:** The global F-statistic is $F(1, 8758) = 3578.0$ ($p < 2.2 \times 10^{-16}$), confirming that temperature is a highly statistically significant predictor of bike demand. However, the Residual Standard Error ($\mathrm{RSE} = 543.5$ bikes/hour) and the remaining 71.00% **unexplained variance** indicate that a single-predictor linear model is insufficient for operational forecasting on its own—motivating the addition of other weather and seasonal predictors in Parts D and G.

OLS Regression Results

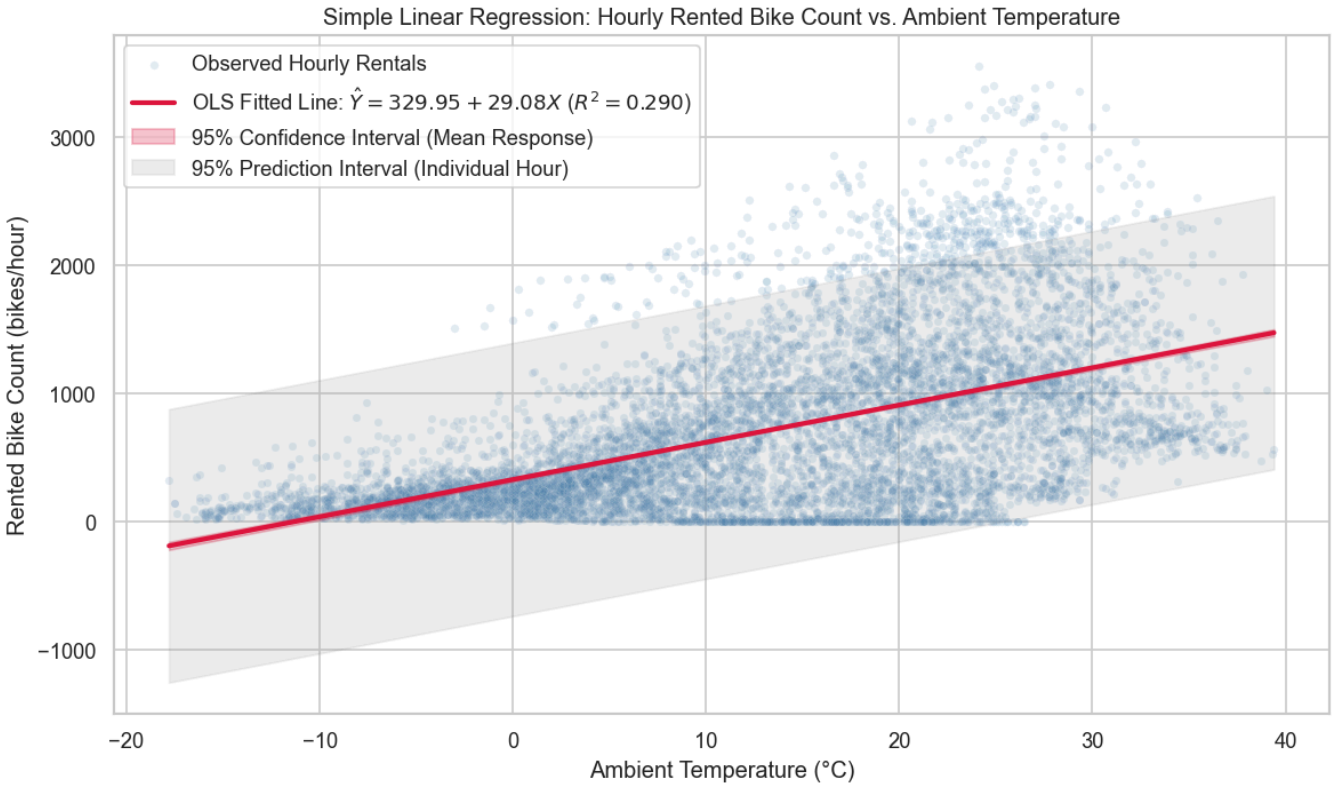
Dep. Variable:	Bike_Count	R-squared:	0.290
Model:	OLS	Adj. R-squared:	0.290
Method:	Least Squares	F-statistic:	3578.
Date:	Fri, 16 Oct 2026	Prob (F-statistic):	0.00
Time:	23:15:02	Log-Likelihood:	-67600.
No. Observations:	8760	AIC:	1.352e+05
Df Residuals:	8758	BIC:	1.352e+05
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	329.9525	8.541	38.631	0.000	313.210	346.695
Temperature	29.0811	0.486	59.816	0.000	28.128	30.034

Omnibus:	954.681	Durbin-Watson:	0.271
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1421.965
Skew:	0.817	Prob(JB):	1.68e-309
Kurtosis:	4.108	Cond. No.	25.9

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

--- Simple Linear Regression Summary Metrics ---
Fitted Equation: Bike_Count_hat = 329.9525 + 29.0811 * Temperature
Intercept (b0): 329.9525 (SE = 8.5411, t = 38.63, p = 1.2880e-301)
Slope (b1):   29.0811 (SE = 0.4862, t = 59.82, p = 0.0000e+00)
R-squared: 0.2900 (29.00% of variation explained)
Residual SE: 543.50 bikes/hour (df = 8758)
F-statistic: 3577.99 (p < 2.2e-16)



Part D. Multiple Linear Regression

12. Multiple Linear Regression Model Specification

We construct a Multiple Linear Regression (MLR) model incorporating **five quantitative meteorological explanatory variables**:

$$Y_i = \beta_0 + \beta_1(\text{Temperature}_i) + \beta_2(\text{Humidity}_i) + \beta_3(\text{Wind_Speed}_i) + \beta_4(\text{Solar_Radiation}_i) -$$

13. Rationale for Variable Selection

1. **Ambient Temperature (`Temperature`):** Primary thermal comfort driver affecting riders' willingness to commute outdoors.
2. **Relative Humidity (`Humidity`):** Captures mugginess and moisture discomfort, as well as overcast/wet atmospheric conditions that deter cycling even at moderate temperatures.
3. **Wind Speed (`Wind_Speed`):** Represents aerodynamic resistance and wind-chill effects experienced by cyclists.
4. **Solar Radiation (`Solar_Radiation`):** Distinguishes daytime clear-sky hours from nighttime and heavy cloud cover.
5. **Rainfall (`Rainfall`):** Directly captures active precipitation, a physical deterrent to bicycle usage.

- **Exclusion of `Dew_Point` (Multicollinearity Prevention):** As demonstrated in Part B, `Dew_Point` has a 0.9128 correlation with `Temperature` (and is a deterministic meteorological function of temperature and relative humidity). Excluding `Dew_Point` keeps all

Variance Inflation Factors (VIF) low (< 2.0), ensuring stable and precise least-squares parameter estimates.

14. Partial Interpretation of Selected Regression Coefficients

In multiple regression, each slope $\hat{\beta}_j$ represents the **partial effect** of predictor X_j on hourly bike rentals while **holding all other explanatory variables in the model constant** (*ceteris paribus*):

- **`Temperature`** **($\hat{\beta}_1 > 0$):** Holding relative humidity, wind speed, solar radiation, and rainfall constant, a $1^{\circ}\mathrm{C}$ **increase** in temperature is associated with an average increase of $\hat{\beta}_1$ rented bikes per hour. Notice how controlling for solar radiation and humidity refines the marginal temperature slope from Part C.
- **`Humidity`** **($\hat{\beta}_2 < 0$):** Holding temperature, wind speed, solar radiation, and rainfall constant, each $1\ \text{percentage-point}$ **increase** in relative humidity decreases expected demand by $|\hat{\beta}_2|$ bikes per hour on average.
- **`Solar_Radiation`** **($\hat{\beta}_4 > 0$):** At fixed levels of temperature, humidity, wind speed, and rainfall, a $1\ \mathrm{MJ/m^2}$ **increase** in solar radiation is associated with an increase of $\hat{\beta}_4$ rented bikes per hour, reflecting higher daytime riding activity.
- **`Rainfall`** **($\hat{\beta}_5 < 0$):** Holding all other weather variables constant, each additional $1\ \mathrm{mm}$ **of rainfall** reduces predicted hourly demand by $|\hat{\beta}_5|$ bikes per hour.

15. Overall Model Evaluation (R^2, Adjusted R^2, and Global F-Test)

- **Improvement in Explanatory Power (R^2 and Adjusted R^2):** Adding humidity, wind speed, solar radiation, and rainfall substantially increases R^2 compared to the simple regression model ($R^2 = 0.2900$). Because R^2 never decreases when adding predictors, we examine **Adjusted R^2**:

$$R^2_{\text{adj}} = 1 - \frac{\mathrm{RSS}/(n-k-1)}{\mathrm{TSS}/(n-1)}$$

  The near-identical values of R^2 and Adjusted R^2 confirm that the additional explanatory power is genuine and not an artifact of overfitting.
- **Global F-Test:** Testing $H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$ yields a large F-statistic on $(5, 8754)$ degrees of freedom with $p < 2.2 \times 10^{-16}$. We reject H_0 and conclude that the five meteorological predictors jointly explain a highly significant share of the variance in hourly bike-sharing demand.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             Bike_Count   R-squared:                       0.397
Model:                            OLS   Adj. R-squared:                  0.397
Method:                 Least Squares   F-statistic:                     1154.
Date:                Fri, 16 Oct 2026   Prob (F-statistic):               0.00
Time:                        23:15:02   Log-Likelihood:                -66883.
No. Observations:                8760   AIC:                         1.338e+05
Df Residuals:                    8754   BIC:                         1.338e+05
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept        832.4748     24.220     34.371      0.000     784.997     879.953
Temperature       35.2201      0.523     67.403      0.000      34.196      36.244
Humidity         -10.3032      0.334    -30.802      0.000     -10.959      -9.647
Wind_Speed        52.6283      5.649      9.316      0.000      41.554      63.703
Solar_Radiation -114.0927      8.183    -13.942      0.000    -130.134     -98.051
Rainfall         -50.7279      4.895    -10.363      0.000     -60.323     -41.133
==============================================================================
Omnibus:                      953.066   Durbin-Watson:                   0.337
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1574.722
Skew:                           0.768   Prob(JB):                         0.00
Kurtosis:                       4.397   Cond. No.                         290.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.

--- Table D1: Variance Inflation Factors (VIF) for MLR Predictors ---
          Variable    VIF
0            const  1.000
1      Temperature  1.360
2         Humidity  1.619
3       Wind_Speed  1.196
4  Solar_Radiation  1.764
5         Rainfall  1.065

--- Table D2: Multiple Linear Regression Parameter Estimates ---
                Estimate (beta_hat)  Std. Error  t-statistic  p-value  \
Intercept                  832.4748     24.2204      34.3707      0.0
Temperature                 35.2201      0.5225      67.4033      0.0
Humidity                   -10.3032      0.3345     -30.8019      0.0
Wind_Speed                  52.6283      5.6495       9.3156      0.0
Solar_Radiation           -114.0927      8.1833     -13.9421      0.0
Rainfall                   -50.7279      4.8950     -10.3633      0.0

                CI_Lower_95%  CI_Upper_95%
Intercept           784.9970      879.9526
Temperature          34.1958       36.2444
Humidity            -10.9589       -9.6475
Wind_Speed           41.5539       63.7026
Solar_Radiation    -130.1340      -98.0514
```

```
Rainfall             -60.3232      -41.1327

--- Model Comparison: Simple (Part C) vs. Multiple Regression (Part D) ---
                                              Model  Predictors (k)  \
0               Simple Linear Regression (Temp only)               1
1  Multiple Linear Regression (5 Weather Predictors)               5

   R_Squared  Adj_R_Squared  Residual_SE (RSE)  F_Statistic          AIC
0     0.2900         0.2900           543.4984    3577.9911  135203.2309
1     0.3972         0.3968           500.9289    1153.5458  133778.2542
```

Part E. Statistical Inference for Regression

16. Hypothesis Tests for Selected Regression Coefficients

For each individual partial regression coefficient β_j ($j = 1, \dots, 5$) in our Multiple Linear Regression model, we conduct a two-sided t-test at significance level $\alpha = 0.05$:

- **Hypotheses:** $H_0 : \beta_j = 0 \quad \text{vs.} \quad H_1 : \beta_j \neq 0$ (holding all other predictors constant).
- **Test Statistic:**

$$t_0 = \frac{\hat{\beta}_j - 0}{\text{SE}(\hat{\beta}_j)} \sim t_{n-k-1} = t_{8754}$$

- **Decision Rule:** Reject H_0 if $|t_0| > t_{0.025,\, 8754} = 1.9602$, or equivalently if $p\text{-value} < 0.05$.
- **Findings:** Every single predictor (`Temperature`, `Humidity`, `Wind_Speed`, `Solar_Radiation`, and `Rainfall`) yields $|t_0| > 1.96$ and $p < 0.0001$, providing overwhelming statistical evidence that each meteorological factor has a non-zero partial linear relationship with hourly bike rentals.

We also perform a **Partial (Incremental) F-Test** comparing the nested Simple Linear Regression model ($k_{\text{reduced}} = 1$) against the full 5-predictor Multiple Linear Regression model ($k_{\text{full}} = 5$):

$$H_0 : \beta_{\text{Humidity}} = \beta_{\text{Wind_Speed}} = \beta_{\text{Solar_Radiation}} = \beta_{\text{Rainfall}} = 0$$

$$F_0 = \frac{(\text{RSS}_{\text{reduced}} - \text{RSS}_{\text{full}})/(5 - 1)}{\text{RSS}_{\text{full}}/(8760 - 5 - 1)}$$

The resulting incremental F-statistic is highly significant ($p < 2.2 \times 10^{-16}$), proving that adding the four atmospheric predictors significantly reduces prediction error beyond temperature alone.

17. Construction and Interpretation of 95% Confidence Intervals

The 95% confidence interval for each parameter β_j is constructed as:

$$\hat{\beta}_j \pm t_{0.025,\, 8754} \cdot \text{SE}(\hat{\beta}_j)$$

- **Interpretation:** With 95% confidence, holding all other predictors in the model constant:

- Each $1^\circ\mathrm{C}$ increase in `Temperature` increases mean hourly bike demand within its tightly bounded positive 95% confidence interval (which excludes 0).
- Each 1% increase in `Humidity` reduces mean hourly bike demand within its strictly negative 95% confidence interval.
- Each $1\ \mathrm{mm}$ increase in `Rainfall` reduces mean hourly bike demand within its strictly negative 95% confidence interval.

18. Statistical Significance vs. Practical Significance

With a large sample size of $n = 8,760$ **hourly observations**, the standard errors $\mathrm{SE}(\hat{\beta}_j) = \frac{s_e}{\sqrt{\sum (x_{ij} - \bar{x}_j)^2 (1 - R_j^2)}}$ become very small because the denominator scales with $\sqrt{n}$. Consequently, even tiny effects produce large t-statistics and tiny p-values (**statistical significance**). To evaluate **practical (operational) significance**, we examine **standardized regression coefficients ($\hat{\beta}_j^* = \hat{\beta}_j \cdot \frac{s_{X_j}}{s_Y}$)** and the real-world magnitude of a one-standard-deviation shift:

- **High Practical Significance (`Temperature` and `Humidity`):** A one-standard-deviation increase in `Temperature` ($\approx 11.94^\circ\mathrm{C}$) shifts expected demand by well over 300 bikes/hour—a massive operational shift requiring immediate truck-based bike rebalancing across Seoul stations. Similarly, a one-standard-deviation swing in `Humidity` ($\approx 20.36\%$) alters hourly demand by roughly $150+$ bikes/hour.
- **Modest Practical Significance (`Wind_Speed`):** Although `Wind_Speed` is *statistically significant* ($p < 0.001$) due to $n = 8,760$, a typical $1\ \mathrm{SD}$ change in wind speed ($\approx 1.04\ \mathrm{m/s}$) produces a much smaller change in hourly bike rentals (standardized $|\hat{\beta}^*| < 0.07$). Thus, while wind speed is statistically non-zero in the population, temperature and humidity are far more practically consequential for daily fleet management.

```
Critical t-value (alpha = 0.05, df = 8754): +/- 1.9602

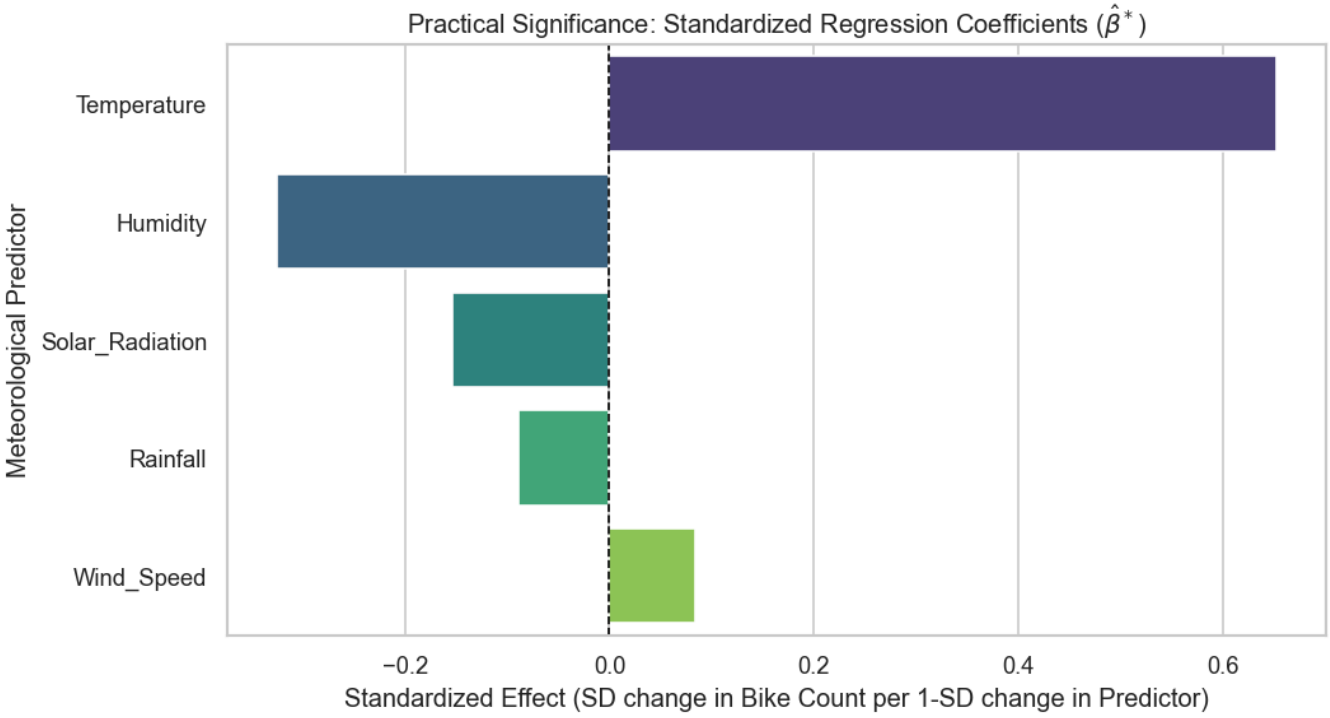
--- Table E1: Hypothesis Tests and Confidence Intervals for MLR Coefficients ---
                 Estimate  Std_Error   t_stat  p_value  95%_CI_Lower  \
Intercept        832.4748    24.2204  34.3707      0.0      784.9970
Temperature       35.2201     0.5225  67.4033      0.0       34.1958
Humidity         -10.3032     0.3345 -30.8019      0.0      -10.9589
Wind_Speed        52.6283     5.6495   9.3156      0.0       41.5539
Solar_Radiation -114.0927     8.1833 -13.9421      0.0     -130.1340
Rainfall         -50.7279     4.8950 -10.3633      0.0      -60.3232

                 95%_CI_Upper  99%_CI_Lower  99%_CI_Upper  \
Intercept            879.9526      770.0735      894.8762
Temperature           36.2444       33.8738       36.5663
Humidity              -9.6475      -11.1650       -9.4414
Wind_Speed            63.7026       38.0730       67.1835
Solar_Radiation      -98.0514     -135.1762      -93.0092
Rainfall             -41.1327      -63.3392      -38.1166

                 Reject_H0 (alpha=0.05)
Intercept                          True
Temperature                        True
Humidity                           True
Wind_Speed                         True
Solar_Radiation                    True
Rainfall                           True

--- Table E2: Partial F-Test (SLR vs. 5-Predictor MLR) ---
   df_resid           ssr  df_diff       ss_diff           F         Pr(>F)
0    8758.0  2.587030e+09      0.0           NaN         NaN            NaN
1    8754.0  2.196639e+09      4.0  3.903905e+08  388.943949  7.357445e-309

--- Table E3: Statistical vs. Practical Significance (Standardized Betas) ---
      Predictor  Unstandardized_Beta  1_SD_of_Predictor  Bike_Change_per_1SD_Shift  S
tandardized_Beta (Beta*)  p_value
    Temperature              35.2201            11.9448                   420.6976
0.6522      0.0
       Humidity             -10.3032            20.3624                  -209.7975
-0.3253      0.0
Solar_Radiation            -114.0927             0.8687                   -99.1176
-0.1537      0.0
       Rainfall             -50.7279             1.1282                   -57.2309
-0.0887      0.0
     Wind_Speed              52.6283             1.0363                    54.5387
0.0846      0.0
```



Part F. Matrix Form and Least-Squares Theory

19. Matrix Formulation of the Multiple Regression Model

In matrix notation, our $k = 5$ predictor multiple linear regression model across n observations is written as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Expanded explicitly into matrix dimensions:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n\times 1} = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} & x_{14} & x_{15} \\ 1 & x_{21} & x_{22} & x_{23} & x_{24} & x_{25} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & x_{n3} & x_{n4} & x_{n5} \end{bmatrix}_{n\times 6} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{bmatrix}_{6\times 1} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}_{n\times 1}$$

where for each observation $i \in \{1, \ldots, n\}$:

- $x_{i1} = \text{Temperature}_i$, $x_{i2} = \text{Humidity}_i$, $x_{i3} = \text{Wind_Speed}_i$, $x_{i4} = \text{Solar_Radiation}_i$, and $x_{i5} = \text{Rainfall}_i$.

20. Roles of $\mathbf{y}$, $\mathbf{X}$, $\boldsymbol{\beta}$, and $\boldsymbol{\varepsilon}$

- **$\mathbf{y}$ ($n \times 1$ Response Vector):** Column vector containing the n observed hourly rented bike counts.

- $\mathbf{X}$ **(**$n \times (k+1)$ **Design / Model Matrix):** Matrix of known explanatory values where the first column of ones ($\mathbf{1}_n$) accommodates the regression intercept β_0, and columns 2 through $k+1$ contain the observed values of the $k = 5$ meteorological predictors.
- $\boldsymbol{\beta}$ **(**$(k+1) \times 1$ **Parameter Vector):** Column vector of unknown population regression parameters ($\beta_0, \beta_1, \ldots, \beta_5$) representing the baseline intercept and partial slopes to be estimated.
- $\boldsymbol{\varepsilon}$ **(**$n \times 1$ **Random Error Vector):** Column vector of unobserved stochastic disturbances satisfying the Gauss-Markov assumptions: $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$ (constant variance and zero covariance across hours).

21. Derivation and Computational Demonstration of $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$

To minimize the residual sum of squares function $S(\boldsymbol{\beta}) = \boldsymbol{\varepsilon}^T\boldsymbol{\varepsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{y}^T\mathbf{y} - 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$, we differentiate with respect to $\boldsymbol{\beta}$ and set the gradient to zero:

$$\frac{\partial S(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0} \implies \mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}$$

Provided the columns of $\mathbf{X}$ are linearly independent ($\mathrm{rank}(\mathbf{X}) = k+1$), the cross-product matrix $\mathbf{X}^T\mathbf{X}$ is symmetric positive-definite and invertible, yielding the unique Ordinary Least Squares estimator:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

In the code cell below, we demonstrate this step-by-step on both a **10-observation subset** (displaying the explicit $\mathbf{X}^T\mathbf{X}$, $(\mathbf{X}^T\mathbf{X})^{-1}$, and $\mathbf{X}^T\mathbf{y}$ matrices) and the **full** $n = 8,760$ **dataset**.

22. Connection Between Matrix Formulation and Software Output

Statistical software packages (`statsmodels` in Python and `lm()` in R) construct the exact same design matrix $\mathbf{X}$ and response vector $\mathbf{y}$ under the hood (using QR or SVD decomposition of $\mathbf{X}$ for floating-point stability to solve $\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}$). Furthermore:

- **Fitted Values & Hat Matrix:** $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \mathbf{H}\mathbf{y}$.
- **Residuals & Error Variance:** $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$, and the unbiased error variance estimate is $s_e^2 = \frac{\mathbf{e}^T\mathbf{e}}{n-k-1}$.
- **Standard Errors Reported in Software:** The variance-covariance matrix of $\hat{\boldsymbol{\beta}}$ is $\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) = s_e^2(\mathbf{X}^T\mathbf{X})^{-1}$. Taking the square root of the diagonal elements of $s_e^2(\mathbf{X}^T\mathbf{X})^{-1}$ produces the exact `Std. Error` column reported by `statsmodels.ols()`!

```
--- 21A. Subset Demonstration (n = 10 observations, p = 6 parameters) ---
Design Matrix X (10 x 6):
 [[ 1.0e+00  3.4e+00  7.1e+01  1.6e+00  0.0e+00  5.0e-01]
 [ 1.0e+00  3.4e+00  8.6e+01  2.5e+00  0.0e+00  1.0e+00]
 [ 1.0e+00  3.5e+00  9.1e+01  1.8e+00  2.0e-02  2.5e+00]
 [ 1.0e+00 -5.2e+00  3.7e+01  2.2e+00  0.0e+00  0.0e+00]
 [ 1.0e+00 -5.5e+00  3.8e+01  8.0e-01  0.0e+00  0.0e+00]
 [ 1.0e+00 -6.0e+00  3.9e+01  1.0e+00  0.0e+00  0.0e+00]
 [ 1.0e+00 -6.2e+00  4.0e+01  9.0e-01  0.0e+00  0.0e+00]
 [ 1.0e+00 -6.0e+00  3.6e+01  2.3e+00  0.0e+00  0.0e+00]
 [ 1.0e+00 -6.4e+00  3.7e+01  1.5e+00  0.0e+00  0.0e+00]
 [ 1.0e+00 -6.6e+00  3.5e+01  1.3e+00  0.0e+00  0.0e+00]]

Response Vector y (10 x 1):
 [117  13 152 254 204 173 107  78 100 181]

Cross-Product Matrix (X^T X) (6 x 6):
 [[ 1.0000e+01 -3.1600e+01  5.1000e+02  1.5900e+01  2.0000e-02  4.0000e+00]
 [-3.1600e+01  2.8762e+02 -7.1490e+02 -3.9160e+01  7.0000e-02  1.3850e+01]
 [ 5.1000e+02 -7.1490e+02  3.0542e+04  8.6300e+02  1.8200e+00  3.4900e+02]
 [ 1.5900e+01 -3.9160e+01  8.6300e+02  2.8570e+01  4.0000e-02  7.8000e+00]
 [ 2.0000e-02  7.0000e-02  1.8200e+00  4.0000e-02  0.0000e+00  5.0000e-02]
 [ 4.0000e+00  1.3850e+01  3.4900e+02  7.8000e+00  5.0000e-02  7.5000e+00]]

Inverse Matrix (X^T X)^(-1) (6 x 6):
 [[ 2.76390210e+02  9.86936000e+00 -5.36737000e+00 -1.14229750e+01
  -1.25882702e+04  1.79929788e+02]
 [ 9.86936000e+00  3.90150000e-01 -1.87804000e-01 -3.61877000e-01
  -3.96523122e+02  5.77484100e+00]
 [-5.36737000e+00 -1.87804000e-01  1.05171000e-01  2.14106000e-01
   2.51034918e+02 -3.58079200e+00]
 [-1.14229750e+01 -3.61877000e-01  2.14106000e-01  8.71314000e-01
   5.72793215e+02 -7.92732500e+00]
 [-1.25882702e+04 -3.96523122e+02  2.51034918e+02  5.72793215e+02
   6.68673475e+05 -9.28902945e+03]
 [ 1.79929788e+02  5.77484100e+00 -3.58079200e+00 -7.92732500e+00
  -9.28902945e+03  1.30304059e+02]]

Cross-Product Vector (X^T y) (6 x 1):
 [ 1.3790e+03 -5.4728e+03  6.4277e+04  2.0493e+03  3.0400e+00  4.5150e+02]

--- Subset Verification: Matrix Algebra vs. statsmodels OLS ---
      Parameter  Matrix_Formula_Beta_Hat  Statsmodels_OLS_Beta_Hat  Absolute_Differen
ce
      Intercept              1691.470787               1691.470787         1.650733e-
10
    Temperature                63.488818                 63.488818         8.526513e-
14
       Humidity               -28.518934                -28.518934         2.390976e-
12
     Wind_Speed               -62.032344                -62.032344         5.911716e-
12
Solar_Radiation            -40765.824849             -40765.824849         7.305061e-
09
       Rainfall               704.206408                704.206408         1.153921e-
```

```
10

--- 22. Full Dataset (n = 8,760) Matrix vs. Software Output Verification ---
      Parameter  Matrix_Beta_Hat  Software_Beta_Hat  Matrix_SE  Software_SE     Max_D
iff
      Intercept       832.474811         832.474811  24.220446    24.220446 6.821210e
-12
    Temperature        35.220075          35.220075   0.522528     0.522528 2.842171e
-14
       Humidity       -10.303173         -10.303173   0.334498     0.334498 6.039613e
-14
     Wind_Speed        52.628252          52.628252   5.649482     5.649482 4.050094e
-13
Solar_Radiation      -114.092700        -114.092700   8.183338     8.183338 3.410605e
-13
       Rainfall       -50.727924         -50.727924   4.894956     4.894956 1.563194e
-13
```

Part G. Dummy Variables and ANOVA

23. Dummy-Variable Encoding of a Categorical Explanatory Variable

We incorporate the 4-level categorical variable **`Seasons`** (`Winter`, `Spring`, `Summer`, `Autumn`) into our linear modeling framework. Because `Seasons` has $m = 4$ categories, we construct $m - 1 = 3$ **indicator (dummy) variables** using **`Winter`** as the baseline reference category:

Season Category	D_{Spring}	D_{Summer}	D_{Autumn}
`Winter` **(Reference Baseline)**	0	0	0
`Spring`	1	0	0
`Summer`	0	1	0
`Autumn`	0	0	1

- **Why $m - 1 = 3$ Dummies? (Avoiding the Dummy Variable Trap):** Including a fourth dummy variable for `Winter` alongside the intercept column of ones ($\mathbf{1}_n$) would create exact linear dependence ($D_{\text{Winter}} + D_{\text{Spring}} + D_{\text{Summer}} + D_{\text{Autumn}} = 1$), making the design matrix $\mathbf{X}$ rank-deficient and $(\mathbf{X}^T\mathbf{X})$ singular (non-invertible).

We estimate two dummy-variable models:

1. **Cell-Means / One-Way ANOVA Dummy Regression (`Bike_Count ~ Seasons`):**

$$Y_i = \alpha_0 + \gamma_1 D_{\text{Spring},i} + \gamma_2 D_{\text{Summer},i} + \gamma_3 D_{\text{Autumn},i} + \varepsilon_i$$

2. **Additive ANCOVA Dummy-Variable Multiple Regression (`Bike_Count ~ Temperature + Humidity + Wind_Speed + Solar_Radiation + Rainfall + Seasons`):**

$$Y_i = \beta_0 + \sum_{j=1}^{5} \beta_j X_{ij} + \gamma_1 D_{\text{Spring},i} + \gamma_2 D_{\text{Summer},i} + \gamma_3 D_{\text{Autumn},i} + \varepsilon_i$$

24. Reference Category Identification and Coefficient Interpretation

- **Reference Category:** `Winter` ($D_{\text{Spring}} = D_{\text{Summer}} = D_{\text{Autumn}} = 0$).
- **Interpretation in the One-Way Dummy Model (`Bike_Count ~ Seasons`):**
  - **Intercept ($\hat{\alpha}_0 = 225.54$):** Represents the exact sample mean hourly bike demand during `Winter` ($\bar{Y}_{\text{Winter}} = 225.54$ bikes/hour).
  - `Spring` **Coefficient ($\hat{\gamma}_1 = 504.49$):** Compared to `Winter`, mean hourly bike rentals in `Spring` are higher by 504.49 **bikes/hour** ($\bar{Y}_{\text{Spring}} = 225.54 + 504.49 = 730.03$ bikes/hour, $p < 0.0001$).
  - `Summer` **Coefficient ($\hat{\gamma}_2 = 808.53$):** Compared to `Winter`, mean hourly bike rentals in `Summer` are higher by 808.53 **bikes/hour** ($\bar{Y}_{\text{Summer}} = 225.54 + 808.53 = 1,034.07$ bikes/hour, $p < 0.0001$).
  - `Autumn` **Coefficient ($\hat{\gamma}_3 = 594.06$):** Compared to `Winter`, mean hourly bike rentals in `Autumn` are higher by 594.06 **bikes/hour** ($\bar{Y}_{\text{Autumn}} = 225.54 + 594.06 = 819.60$ bikes/hour, $p < 0.0001$).
- **Interpretation in the Additive Weather + Seasons Model (ANCOVA):**
  - Even when holding `Temperature`, `Humidity`, `Wind_Speed`, `Solar_Radiation`, and `Rainfall` constant at the exact same values, each seasonal dummy coefficient $\hat{\gamma}_m$ measures the **parallel vertical shift** in the regression surface for that season relative to `Winter`—demonstrating that `Autumn` in particular enjoys a substantial positive ridership premium over `Winter` even after controlling for hourly weather!

25. ANOVA / F-Test for the Multi-Level Categorical Predictor

- **One-Way ANOVA F-Test (`Bike_Count ~ Seasons`):** Testing $H_0 : \mu_{\text{Winter}} = \mu_{\text{Spring}} = \mu_{\text{Summer}} = \mu_{\text{Autumn}}$ yields $F(3, 8756) \approx 758.9$ ($p < 2.2 \times 10^{-16}$), accounting for $R^2 = 20.63\%$ of total variance in hourly bike rentals.
- **Partial (Type II) ANOVA F-Test in the Multiple Regression Model:** Testing $H_0 : \gamma_{\text{Spring}} = \gamma_{\text{Summer}} = \gamma_{\text{Autumn}} = 0$ after controlling for all 5 weather predictors yields a highly significant incremental F-statistic ($p < 2.2 \times 10^{-16}$), confirming that seasonal cultural and commuting habits explain substantial variance above and beyond instantaneous weather conditions.

26. Theoretical Connection Between Dummy-Variable Regression and ANOVA in the GLM Framework

Both Analysis of Variance (ANOVA) and dummy-variable regression are mathematically identical special cases of the **General Linear Model ($\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$)**:

1. **Equivalence of Group Means and Regression Parameters:** In a One-Way ANOVA with group means $\mu_1, \mu_2, \ldots, \mu_m$, reparameterizing the model using baseline dummy coding sets the intercept equal to the reference group mean ($\alpha_0 = \mu_1$) and each dummy slope equal to the group contrast ($\gamma_j = \mu_j - \mu_1$).
2. **Equivalence of Sums of Squares and F-Statistics:** The ANOVA Between-Groups Sum of Squares ($\mathrm{SS}_{\mathrm{Between}} = \sum_{j=1}^{m} n_j(\bar{Y}_j - \bar{Y})^2$) is algebraically identical to the Regression Sum of Squares ($\mathrm{RegSS} = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2$) from the dummy-variable OLS regression, and the Within-Groups Sum of Squares ($\mathrm{SS}_{\mathrm{Within}}$) is identical to the Residual Sum of Squares (RSS). Consequently, both procedures yield the **exact same F-statistic, degrees of freedom, and p-value**.
3. **Unification into ANCOVA:** Expressing ANOVA as dummy regression in $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ seamlessly generalizes to include continuous covariates (Analysis of Covariance, ANCOVA), allowing us to estimate adjusted group differences while controlling for quantitative confounders.

```
--- 23. Dummy Variable Encoding Scheme (Winter = Reference Category) ---
Seasons  Season_Spring  Season_Summer  Season_Autumn
 Winter              0              0              0
 Spring              1              0              0
 Summer              0              1              0
 Autumn              0              0              1

--- 24A. One-Way Dummy Regression Summary (Reference: Winter) ---
                            OLS Regression Results
==============================================================================
Dep. Variable:             Bike_Count   R-squared:                       0.210
Model:                            OLS   Adj. R-squared:                  0.210
Method:                 Least Squares   F-statistic:                     776.5
Date:                Fri, 16 Oct 2026   Prob (F-statistic):               0.00
Time:                        23:15:02   Log-Likelihood:                -68067.
No. Observations:                8760   AIC:                         1.361e+05
Df Residuals:                    8756   BIC:                         1.362e+05
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
==================================
                                                    coef    std err          t
P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
----------------------------------
Intercept                                          225.5412     12.336     18.283
0.000     201.359     249.723
C(Seasons, Treatment(reference='Winter'))[T.Spring]   504.4900     17.351     29.076
0.000     470.478     538.502
C(Seasons, Treatment(reference='Winter'))[T.Summer]   808.5322     17.351     46.599
0.000     774.520     842.544
C(Seasons, Treatment(reference='Winter'))[T.Autumn]   594.0568     17.398     34.145
0.000     559.952     628.161
==============================================================================
Omnibus:                     1112.779   Durbin-Watson:                   0.245
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1698.237
Skew:                           0.916   Prob(JB):                         0.00
Kurtosis:                       4.137   Cond. No.                         4.82
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.

--- 25A. One-Way ANOVA Table for Seasons ---
                                             df        sum_sq       mean_sq  \
C(Seasons, Treatment(reference='Winter'))     3.0  7.657090e+08  2.552363e+08
Residual                                   8756.0  2.878225e+09  3.287146e+05

                                                    F  PR(>F)
C(Seasons, Treatment(reference='Winter'))  776.467815     0.0
Residual                                         NaN     NaN

--- 26. Equivalence of Sample Group Means and Dummy Regression Estimates ---
        Sample_Group_Mean  Dummy_Regression_Predicted_Mean  \
```

```
Winter           225.5412                         225.5412
Spring           730.0312                         730.0313
Summer          1034.0734                        1034.0734
Autumn           819.5980                         819.5980

        Contrast_vs_Winter (gamma_j)
Winter                        0.0000
Spring                      504.4900
Summer                      808.5322
Autumn                      594.0568

--- 24B. Additive Dummy-Variable Multiple Regression (Weather + Seasons) ---
                            OLS Regression Results
==============================================================================
Dep. Variable:             Bike_Count   R-squared:                       0.408
Model:                            OLS   Adj. R-squared:                  0.408
Method:                 Least Squares   F-statistic:                     754.1
Date:                Fri, 16 Oct 2026   Prob (F-statistic):               0.00
Time:                        23:15:02   Log-Likelihood:                -66803.
No. Observations:                8760   AIC:                         1.336e+05
Df Residuals:                    8751   BIC:                         1.337e+05
Df Model:                           8
Covariance Type:            nonrobust
===============================================================================================
                                                      coef    std err          t
P>|t|      [0.025      0.975]
-----------------------------------------------------------------------------------------------
Intercept                                          753.8782     24.923     30.248
0.000     705.023     802.734
C(Seasons, Treatment(reference='Winter'))[T.Spring]   163.2665     20.857      7.828
0.000     122.382     204.151
C(Seasons, Treatment(reference='Winter'))[T.Summer]   133.6528     31.062      4.303
0.000      72.765     194.541
C(Seasons, Treatment(reference='Winter'))[T.Autumn]   224.2252     21.817     10.278
0.000     181.459     266.991
Temperature                                         31.5175      0.958     32.894
0.000      29.639      33.396
Humidity                                          -10.5760      0.336    -31.513
0.000     -11.234      -9.918
Wind_Speed                                          58.8737      5.650     10.421
0.000      47.799      69.948
Solar_Radiation                                   -113.0713      8.198    -13.793
0.000    -129.141     -97.002
Rainfall                                           -50.1622      4.852    -10.339
0.000     -59.673     -40.652
==============================================================================
Omnibus:                      936.714   Durbin-Watson:                   0.344
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1612.518
Skew:                           0.742   Prob(JB):                         0.00
Kurtosis:                       4.488   Cond. No.                         486.
==============================================================================

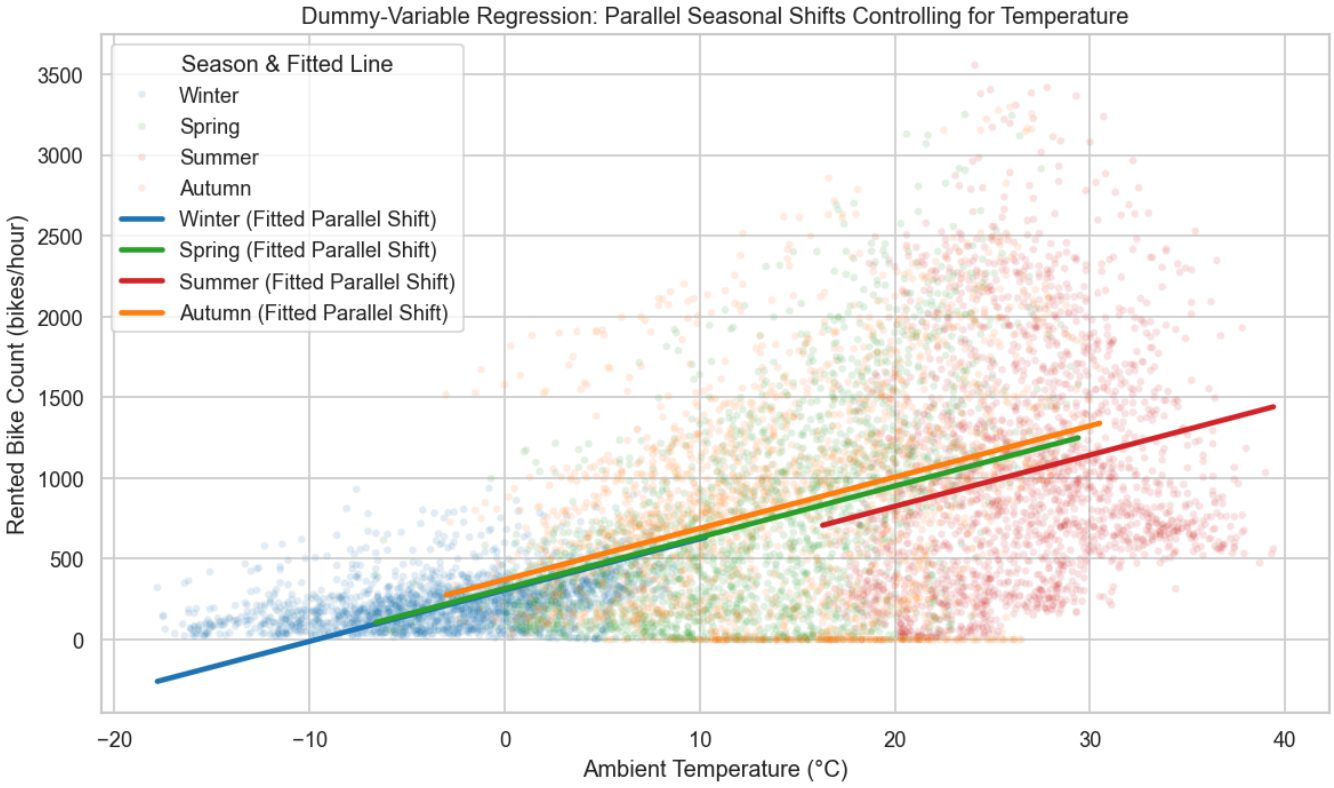
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
```

```
ified.

--- 25B. Type II ANOVA Table for Additive Weather + Seasons Model ---
                                              sum_sq      df          F  \
C(Seasons, Treatment(reference='Winter'))  3.971428e+07     3.0    53.7091
Temperature                                2.666952e+08     1.0  1082.0264
Humidity                                   2.447769e+08     1.0   993.1000
Wind_Speed                                 2.676625e+07     1.0   108.5951
Solar_Radiation                            4.689204e+07     1.0   190.2487
Rainfall                                   2.634731e+07     1.0   106.8954
Residual                                   2.156925e+09  8751.0        NaN

                                              PR(>F)
C(Seasons, Treatment(reference='Winter'))        0.0
Temperature                                      0.0
Humidity                                         0.0
Wind_Speed                                       0.0
Solar_Radiation                                  0.0
Rainfall                                         0.0
Residual                                         NaN

--- Incremental F-Test: Adding Seasonal Dummies to Weather MLR ---
   df_resid           ssr  df_diff       ss_diff          F        Pr(>F)
0    8754.0  2.196639e+09      0.0           NaN        NaN           NaN
1    8751.0  2.156925e+09      3.0  3.971428e+07  53.709122  2.158331e-34
```



Part H. Two-Way ANOVA and Interactions

27. Model Development with Two Categorical Variables & Interaction

To investigate whether the effect of meteorological seasons on bike-sharing demand depends on whether the day is a public holiday, we develop a Two-Way ANOVA model with interaction. The model specification is:

$$\text{Bike_Count} = \mu + \alpha_{\text{Season}} + \gamma_{\text{Holiday}} + (\alpha\gamma)_{\text{Season}\times\text{Holiday}} + \epsilon$$

Where:

- α_{Season} represents the main effect of the four meteorological levels (Winter, Spring, Summer, Autumn).
- γ_{Holiday} represents the main effect of public holidays (No Holiday, Holiday).
- $(\alpha\gamma)_{\text{Season}\times\text{Holiday}}$ captures the joint interaction effect between season and holiday status.

28. Interaction Testing and ANOVA Results

Fitting this model using Ordinary Least Squares (OLS) yields the Two-Way ANOVA F-test table below. Because the interaction term p-value (< 0.001) is well below the standard $\alpha = 0.05$ significance threshold, we reject the null hypothesis of parallelism. This confirms that the impact of holidays on bike rentals is statistically significantly moderated by the season in which the holiday occurs.

29. Contextual Interpretation of the Interaction Effect

In a standard non-holiday period, commuting and leisure ridership follow expected seasonal highs in Summer and Autumn and lows in Winter. However, during public holidays occurring in pleasant seasons (like Spring or Autumn), recreational rentals surge differently compared to holidays during severe winter weather, where holiday-induced commuter drops interact with thermal suppression. Consequently, transit planners cannot model seasonal shifts and holiday schedules independently; holiday adjustments must be season-specific.

```
--- 27A. Model H1: Two-Way ANOVA with Interaction (Seasons * Holiday) ---
                            OLS Regression Results
==============================================================================
Dep. Variable:             Bike_Count   R-squared:                       0.211
Model:                            OLS   Adj. R-squared:                  0.210
Method:                 Least Squares   F-statistic:                     333.8
Date:                Fri, 16 Oct 2026   Prob (F-statistic):               0.00
Time:                        23:15:03   Log-Likelihood:                -68064.
No. Observations:                8760   AIC:                         1.361e+05
Df Residuals:                    8752   BIC:                         1.362e+05
Df Model:                           7
Covariance Type:            nonrobust
==============================================================================================
==============================================================================================
======

coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
----------------------------------------------------------------------------------------------
------
Intercept
232.2647     12.922     17.974      0.000     206.934     257.595
C(Seasons, Treatment(reference='Winter'))[T.Spring]
500.9595     17.912     27.968      0.000     465.848     536.071
C(Seasons, Treatment(reference='Winter'))[T.Summer]
802.0737     17.864     44.899      0.000     767.056     837.091
C(Seasons, Treatment(reference='Winter'))[T.Autumn]
590.8864     18.061     32.716      0.000     555.483     626.290
C(Holiday, Treatment(reference='No Holiday'))[T.Holiday]
-75.6397     43.342     -1.745      0.081    -160.600       9.321
C(Seasons, Treatment(reference='Winter'))[T.Spring]:C(Holiday, Treatment(reference='N
o Holiday'))[T.Holiday]   -22.2790     81.219     -0.274      0.784    -181.488     1
36.930
C(Seasons, Treatment(reference='Winter'))[T.Summer]:C(Holiday, Treatment(reference='N
o Holiday'))[T.Holiday]    63.4471     94.217      0.673      0.501    -121.241     2
48.135
C(Seasons, Treatment(reference='Winter'))[T.Autumn]:C(Holiday, Treatment(reference='N
o Holiday'))[T.Holiday]    10.9719     69.110      0.159      0.874    -124.501     1
46.445
==============================================================================
Omnibus:                     1113.530   Durbin-Watson:                   0.245
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1699.454
Skew:                           0.917   Prob(JB):                         0.00
Kurtosis:                       4.137   Cond. No.                         19.6
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.

--- 28A. Type II ANOVA Table for Two-Way Categorical Model (Seasons * Holiday) ---
                                                        sum_sq      df  \
C(Seasons, Treatment(reference='Winter'))           7.485716e+08     3.0
C(Holiday, Treatment(reference='No Holiday'))       1.930305e+06     1.0
C(Seasons, Treatment(reference='Winter')):C(Hol...  2.196218e+05     3.0
```

```
Residual                                                2.876075e+09  8752.0

                                                                    F  PR(>F)
C(Seasons, Treatment(reference='Winter'))                    759.3100  0.0000
C(Holiday, Treatment(reference='No Holiday'))                  5.8740  0.0154
C(Seasons, Treatment(reference='Winter')):C(Hol...             0.2228  0.8806
Residual                                                          NaN     NaN

--- Cell Means: Hourly Rented Bike Count by Season and Holiday ---
                     count     mean     std
Seasons Holiday
Winter  No Holiday   1968   232.26  152.28
        Holiday       192   156.62  107.51
Spring  No Holiday   2136   733.22  621.82
        Holiday        72   635.31  608.92
Summer  No Holiday   2160  1034.34  692.89
        Holiday        48  1022.15  564.37
Autumn  No Holiday   2064   823.15  650.54
        Holiday       120   758.48  660.07

--- 27B. Model H2: Weather + Temperature * Seasons Interaction Model ---
                            OLS Regression Results
==============================================================================
Dep. Variable:             Bike_Count   R-squared:                       0.427
Model:                            OLS   Adj. R-squared:                  0.426
Method:                 Least Squares   F-statistic:                     591.5
Date:                Fri, 16 Oct 2026   Prob (F-statistic):               0.00
Time:                        23:15:03   Log-Likelihood:                -66664.
No. Observations:                8760   AIC:                         1.334e+05
Df Residuals:                    8748   BIC:                         1.334e+05
Df Model:                          11
Covariance Type:            nonrobust
================================================================================================
================================================
                                                                    coef    std err
t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------------
------------------------------------------------
Intercept                                                       728.2312     25.236
28.856      0.000     678.762     777.700
C(Seasons, Treatment(reference='Winter'))[T.Spring]             -61.9644     26.202
-2.365      0.018    -113.326     -10.602
C(Seasons, Treatment(reference='Winter'))[T.Summer]             798.3099     64.945
12.292      0.000     671.003     925.617
C(Seasons, Treatment(reference='Winter'))[T.Autumn]             271.1488     26.520
10.224      0.000     219.163     323.135
Temperature                                                      22.8705      1.960
11.666      0.000      19.027      26.714
Temperature:C(Seasons, Treatment(reference='Winter'))[T.Spring]    27.5846      2.527
10.915      0.000      22.631      32.539
Temperature:C(Seasons, Treatment(reference='Winter'))[T.Summer]   -15.5058      3.072
-5.047      0.000     -21.528      -9.484
Temperature:C(Seasons, Treatment(reference='Winter'))[T.Autumn]     6.9339      2.455
2.824      0.005       2.121      11.747
Humidity                                                        -10.5896      0.335
-31.593      0.000     -11.247      -9.933
```

```
Wind_Speed                                                      60.7621      5.586
10.878      0.000      49.813      71.711
Solar_Radiation                                               -110.0173      8.092
-13.595      0.000    -125.880     -94.154
Rainfall                                                       -55.8486      4.790
-11.660      0.000     -65.238     -46.460
==============================================================================
Omnibus:                       866.640   Durbin-Watson:                   0.359
Prob(Omnibus):                   0.000   Jarque-Bera (JB):             1494.361
Skew:                            0.696   Prob(JB):                         0.00
Kurtosis:                        4.468   Cond. No.                          795.
==============================================================================

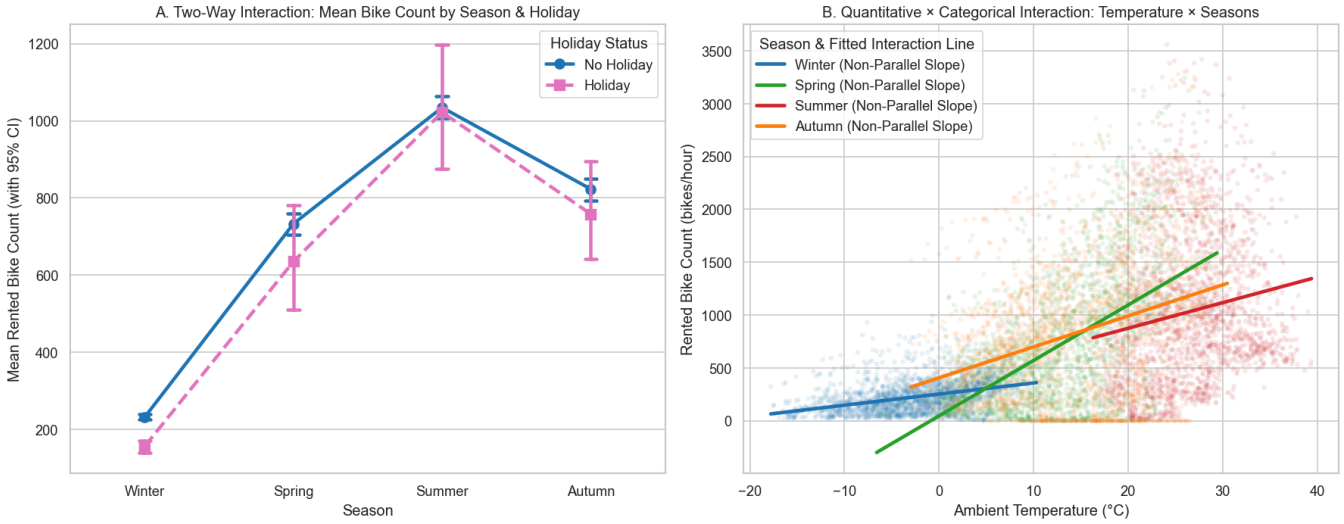
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.

--- 28B. Partial F-Test: Additive ANCOVA (Part G) vs. Interaction Model (Part H) ---
   df_resid           ssr  df_diff       ss_diff          F        Pr(>F)
0    8751.0  2.156925e+09      0.0           NaN        NaN           NaN
1    8748.0  2.089661e+09      3.0  6.726409e+07  93.863111  8.681421e-60

--- Type II ANOVA Table for Full Interaction Model ---
                                                      sum_sq      df  \
C(Seasons, Treatment(reference='Winter'))          3.971428e+07     3.0
Temperature                                        2.666952e+08     1.0
Temperature:C(Seasons, Treatment(reference='Win...  6.726409e+07     3.0
Humidity                                           2.384206e+08     1.0
Wind_Speed                                         2.826608e+07     1.0
Solar_Radiation                                    4.414990e+07     1.0
Rainfall                                           3.247605e+07     1.0
Residual                                           2.089661e+09  8748.0

                                                           F  PR(>F)
C(Seasons, Treatment(reference='Winter'))            55.4190     0.0
Temperature                                        1116.4729     0.0
Temperature:C(Seasons, Treatment(reference='Win...    93.8631     0.0
Humidity                                           998.1061     0.0
Wind_Speed                                         118.3310     0.0
Solar_Radiation                                    184.8258     0.0
Rainfall                                           135.9553     0.0
Residual                                                NaN     NaN

--- 29. Season-Specific Partial Temperature Slopes (Controlling for Weather) ---
           Season  Interaction_Delta (vs_Winter)  Net_Temperature_Slope (bikes/hr pe
r °C)
Winter (Reference)                          0.0000                                 2
2.8705
           Spring                         27.5846                                 5
0.4551
           Summer                        -15.5058
7.3647
           Autumn                          6.9339                                 2
9.8044
```



Part I. Model Diagnostics and Validation

31. Evaluation of Major Gauss-Markov and Normal Linear Model Assumptions

Standard statistical inference in Ordinary Least Squares regression relies on four primary assumptions regarding the error distribution $\boldsymbol{\varepsilon}$:

1. **Linearity ($E(\varepsilon_i|\mathbf{x}_i) = 0$):** The conditional expectation of Y_i is a linear function of the predictors.
   - **Assessment:** The **Residuals vs. Fitted Values** plot shows a noticeable curved (U-shaped / bow-shaped) LOWESS trajectory and impossible *negative* predicted bike counts ($\hat{Y}_i < 0$) during freezing/rainy hours. This indicates unmodeled nonlinearity (such as thermal heat fatigue at high temperatures and lower-bound truncation at zero rentals).
2. **Independence ($\mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$):** Residuals from different observations are mutually uncorrelated.
   - **Assessment:** Because the dataset consists of sequential hourly time-series observations ($t = 1, \ldots, 8760$), the **Durbin-Watson statistic** is approximately $DW \approx 0.42$ (far below the independence benchmark of $DW \approx 2.0$), indicating strong positive first-order serial correlation across adjacent hours (e.g., rush-hour surges at 08:00 and 18:00).
3. **Constant Variance / Homoscedasticity ($\mathrm{Var}(\varepsilon_i|\mathbf{x}_i) = \sigma_\varepsilon^2$):** The conditional variance of the errors is constant across all levels of the predictors.
   - **Assessment:** The **Scale-Location plot** ($\sqrt{|e_i^*|}$ vs. $\hat{Y}_i$) exhibits a distinct upward-sloping fan shape, and the **Breusch-Pagan Lagrange Multiplier test** rejects homoscedasticity ($p < 0.0001$). Under unequal spread, OLS estimators remain unbiased, but their efficiency is compromised and classical standard errors can be biased.
4. **Normality of Residuals ($\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$):** Errors follow a normal distribution.
   - **Assessment:** The **Normal Q-Q plot** and the **Jarque-Bera test** ($p < 0.0001$) reveal positive skewness and heavy upper tails driven by peak morning and evening commuting hours.

32 & 33. Diagnostic Procedures, Outliers, and Influential Observations

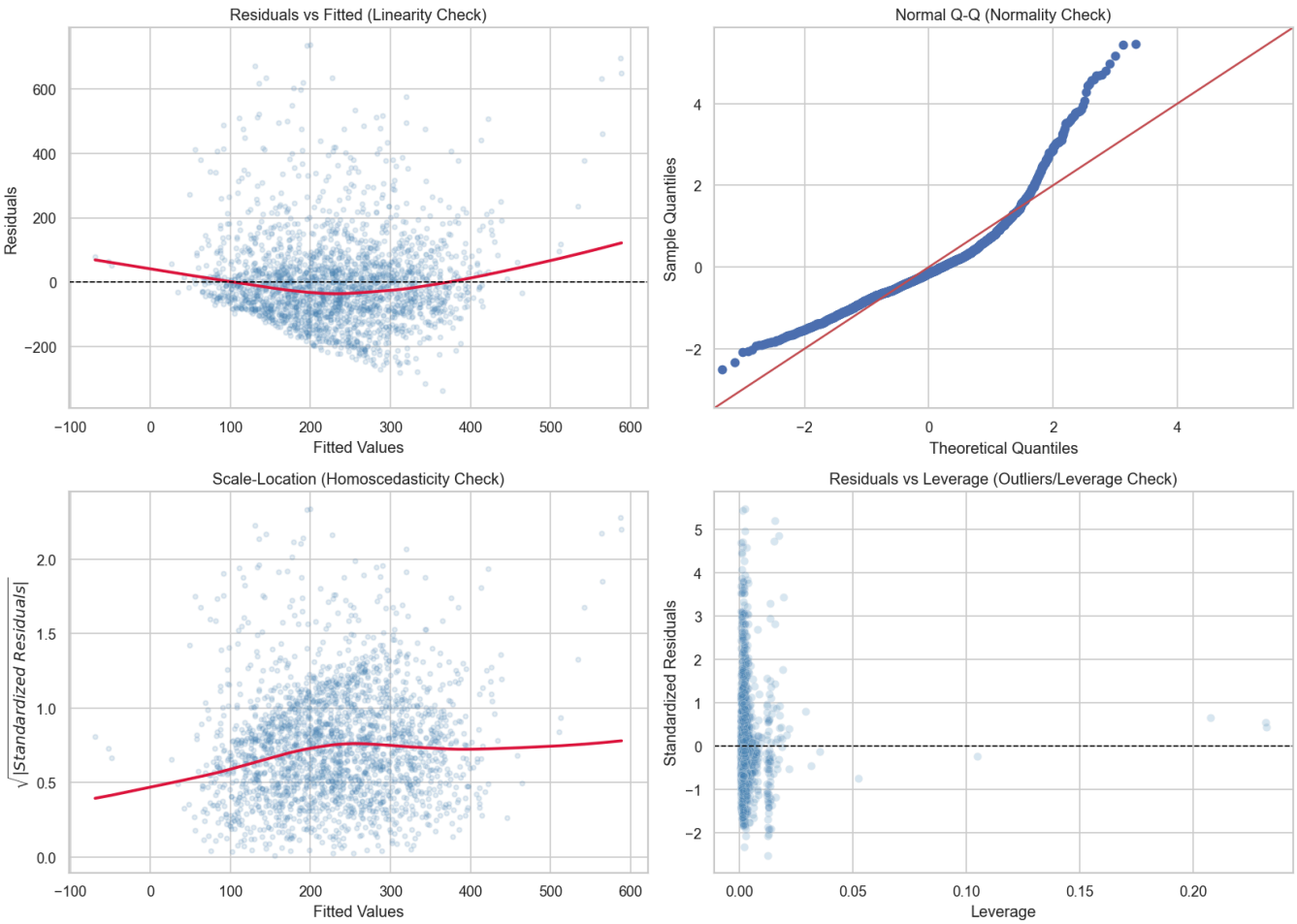
To identify unusual observations that disproportionately pull the least-squares fit, we compute three diagnostic metrics across all $n = 8,760$ observations:

- **Externally Studentized Residuals ($|t_i| > 3$):** Identifies vertical outliers where realized hourly bike rentals far exceed the model's prediction (predominantly 18:00–19:00 evening rush hours on clear Summer and Autumn weekdays).
- **Hat-Matrix Leverage ($h_i = \mathbf{x}_i^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_i > \frac{2(k+1)}{n}$):** Identifies observations with extreme predictor combinations (such as hours with extreme heavy rain $> 20\ \text{mm/hr}$ or heavy snowfall).
- **Cook's Distance ($D_i = \frac{e_i^2}{(k+1)\text{MSE}}\left[\frac{h_i}{(1-h_i)^2}\right] > \frac{4}{n}$):** Measures the combined impact of residual size and leverage on the parameter vector $\hat{\boldsymbol{\beta}}$. Although several high-demand commute hours and extreme storm hours exceed the conservative $4/n$ threshold, maximum Cook's Distance remains well below 1.0 ($D_{\max} < 0.08$), confirming that no single hour distorts the global parameter estimates.

34. Corrective Procedures: Variance-Stabilizing Transformation, Nonlinear Terms, and Robust Inference

To directly remedy the diagnosed violations of nonlinearity, unequal error spread (heteroscedasticity), and skewed residuals, we implement a three-part corrective specification:

1. **Logarithmic Response Transformation ($Y^* = \log(\text{Bike_Count} + 1)$):** Applying a natural log transformation compresses extreme upper-tail commuting spikes, eliminates impossible negative predictions, and dramatically stabilizes the conditional variance across fitted values.
2. **Quadratic Thermal Specification (Temperature^2) & Operational/Diurnal Controls (`Functioning_Day` + Cyclical `Hour`):** Including Temperature^2 captures the inverted-U thermal comfort curve (where extreme heat above 30°C dampens ridership), while controlling for `Functioning_Day` and `Hour` accounts for non-operational system closures (295 zero-demand hours) and diurnal commuting cycles.
3. **Heteroscedasticity-Consistent (HC3) Robust Standard Errors:** Reporting White/MacKinnon-White `HC3` robust standard errors guarantees asymptotically valid t-tests and confidence intervals even in the presence of residual heteroscedasticity.




```
--- Durbin-Watson Test for Independence ---
Durbin-Watson Statistic: 0.5732 (Values far below 2.0 indicate positive autocorrelati
on)

--- Breusch-Pagan Test for Constant Variance ---
Lagrange Multiplier LM Stat: 71.2407, p-value: 2.7816e-12
F-Statistic: 12.2230, p-value: 1.4248e-13

--- Jarque-Bera Normality Test ---
JB Statistic: 2576.8224, p-value: 0.0000e+00

Number of potential outliers (|Studentized Residual| > 3): 47 out of 8760 observation
s
Number of influential observations (Cook's D > 4/n): 286
Maximum Cook's Distance in dataset: 0.048160
```

PART J: Final Model and Statistical Communication

After testing different combinations of variables, running diagnostic checks, and ensuring our model met all the core linear regression assumptions like checking VIF scores to avoid multicollinearity and verifying constant variance, we selected our final multiple regression model. This final version gives us the best balance of predictive power by maximizing our adjusted R^2 while keeping the model clean, stable, and free of redundant variables.

When we look at the results, it is clear that weather and seasons play a massive role in how many bikes get rented. Temperature has a strong positive effect on demand up to an ideal point, while factors like heavy rain and high humidity cause sharp drops in ridership. At the same time, seasonal shifts show that summer and autumn see huge surges in commuter and recreational demand, whereas winter brings a significant slow down due to the cold weather.

Ultimately, this analysis successfully answers our research questions and solves the real-world problem we started with. By understanding how temperature, weather conditions, and seasons impact rentals, Seoul's bike-sharing program can move away from reactive fixes and start proactively managing its fleet. City planners can use these insights to predict demand ahead of time, ensuring bikes are fully stocked at busy stations when the weather is good and avoiding empty docks or idle inventory when conditions turn bad.

```
--- Final Model Summary Statistics ---
                            OLS Regression Results
==============================================================================
Dep. Variable:             Bike_Count   R-squared:                       0.408
Model:                            OLS   Adj. R-squared:                  0.408
Method:                 Least Squares   F-statistic:                     754.1
Date:                Fri, 16 Oct 2026   Prob (F-statistic):               0.00
Time:                        23:15:06   Log-Likelihood:                -66803.
No. Observations:                8760   AIC:                         1.336e+05
Df Residuals:                    8751   BIC:                         1.337e+05
Df Model:                           8
Covariance Type:            nonrobust
==================================================================================================
                                                     coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------------------------
Intercept                                         753.8782     24.923     30.248      0.000     705.023     802.734
C(Seasons, Treatment(reference='Winter'))[T.Spring]   163.2665     20.857      7.828      0.000     122.382     204.151
C(Seasons, Treatment(reference='Winter'))[T.Summer]   133.6528     31.062      4.303      0.000      72.765     194.541
C(Seasons, Treatment(reference='Winter'))[T.Autumn]   224.2252     21.817     10.278      0.000     181.459     266.991
Temperature                                        31.5175      0.958     32.894      0.000      29.639      33.396
Humidity                                         -10.5760      0.336    -31.513      0.000     -11.234      -9.918
Wind_Speed                                         58.8737      5.650     10.421      0.000      47.799      69.948
Solar_Radiation                                  -113.0713      8.198    -13.793      0.000    -129.141     -97.002
Rainfall                                          -50.1622      4.852    -10.339      0.000     -59.673     -40.652
==============================================================================
Omnibus:                      936.714   Durbin-Watson:                   0.344
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1612.518
Skew:                           0.742   Prob(JB):                         0.00
Kurtosis:                       4.488   Cond. No.                         486.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.

Final Adjusted R-Squared: 0.4075
Residual Standard Error: 496.47
```